when the switch is activated. On the other hand, if the circuit were connected to the N.C. and C. contacts, the circuit would remain closed until the switch is activated, when it would open the circuit.

In addition to momentary and lever switches, slide switches or toggle switches stay in whatever position they were last left in. Slide and toggle switches are just like your vanilla, household light switch. Tilt switches, like we used earlier in the book, have a rolling ball inside that close or open the contacts depending on the angle of the switch. Reed switches have a pair of thin metal contacts inside a glass container that will close when a magnet passes near. These are often seen in alarm systems to alert when a window or door has been opened.

Relays have also been included here with other mechanical switches. The relay is a switch that is activated by a magnetic coil. When the coil is given an electric current, it closes the nearby contacts. Where transistors are only good for switching DC voltages, some relays can switch much higher AC voltages, although these relays will often need a transistor to power the larger magnetic coil to activate the switch. Many relays will also have multiple N.O. and N.C. positions available to hook up many different things.

Motors

Motors are included here because they are another common mechanical component used in many of our projects. They will either create radial motion, what we might typically associate with a motor, as well as linear motion using solenoids or other devices. Figure 12-9 shows a few of the many different types of motors available.

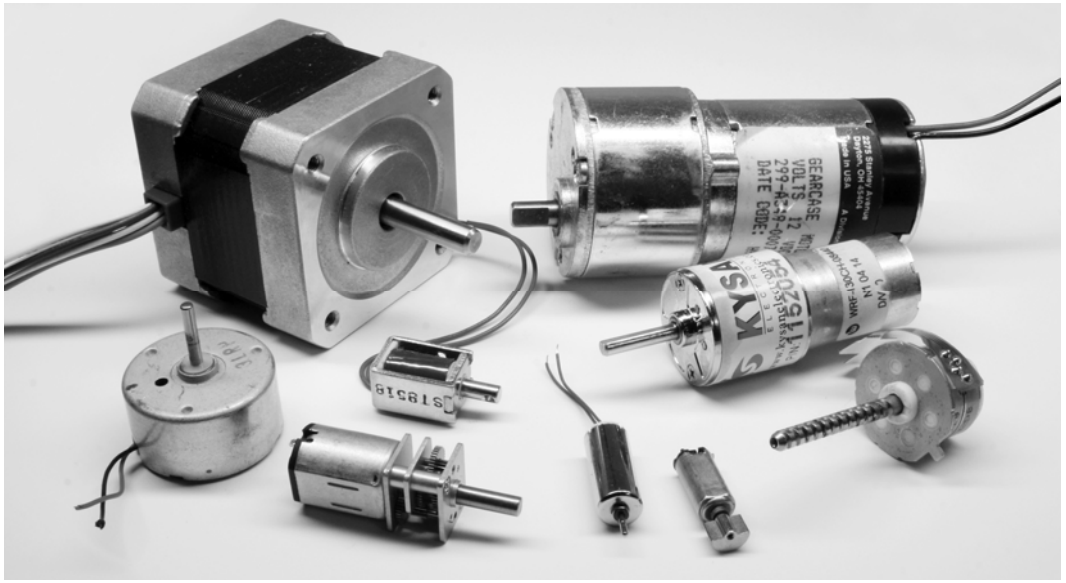


Figure 12-9. *Motors*

These are all DC-type motors. Motors that run on AC are as plentiful, but are also a bit more difficult to interface with microcontrollers. Radial DC motors are offered with or without built-in gearboxes.